

Module 2 Summative Assessment

Instructions

1.

What is a common technique to check if linear regression adequately models the dependent variable?

Ready to review what you've learned before starting the assignment? I'm here to help.

☒ Checking if the assumptions of linear regression are satisfied.

☐ Increasing the regularization parameter to avoid overfitting.

☐ Using only categorical predictors to simplify the interpretation of coefficients.

☐ Adding more predictors to the model until all p -values are significant.

Correct

Due Nov 3, 11:59 PM CST

Attempts 0 left (1 attempt every 8 hours)

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Submitted Nov 3, 3:02 AM CST

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2.

In linear regression, what would adding interaction terms between predictors allow for?

They ensure that the residuals will be normally distributed.

☐ It guarantees that the residuals will be normally distributed.

☒ It allows the model to capture the combined effect of predictors on the dependent variable.

☐ It simplifies the model by reducing the number of predictors.

Correct

100%

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3.

What is the effect of applying a regularization parameter that is too large in a Ridge Regression model?

It will ensure that all predictor coefficients are set to zero.

☐ It will ensure that all predictor coefficients are set to zero.

☒ It may lead to underfitting, where the model is too simplistic to capture the underlying pattern in the data.

☐ It will cause the model to fit perfectly to the training data.

Correct

4.

When is it appropriate to use a data transformation method over ordinary least squares regression?

When the data perfectly fits the assumptions of linear regression.

☐ When the data perfectly fits the assumptions of linear regression.

☐ When the number of predictors is greater than the number of observations.

☒ When the data contain outliers or are heteroskedastic.

☐ When the dependent variable is binary.

Correct

5.

What does the slope in a simple linear regression model represent?

The correlation coefficient between the dependent and independent variables.

☐ The correlation coefficient between the dependent and independent variables.

☐ The intercept of the regression line with the y-axis.

☒ The rate at which the dependent variable changes for each unit change in the independent variable.

☐ The maximum value of the dependent variable.

Correct

6.

Why would a data scientist use a LASSO regression instead of traditional linear regression?

To ensure that the residuals are normally distributed.

☐ To ensure that the residuals are normally distributed.

☐ To guarantee that the model will have a higher R-squared value.

☐ To allow the model to fit non-linear relationships without transformations.

☒ To perform both variable selection and regularization to enhance prediction accuracy.

Correct

7.

When dealing with a cardinal relationship between the independent and dependent variables, which transformation could be appropriate to attempt linearization?

Normal transformation

☐ Normal transformation

☐ Absolute transformation

☐ Binary transformation

☒ Polynomial transformation.

Correct

8.

What is a common method used to select the regularization parameter in Ridge Regression and LASSO?

Cross-validation

☒ Cross-validation

☐ First risk based on the number of predictors.

☐ Expert judgment only

☐ Always setting it to a default value of 1.

Correct

9.

Based on what you know regarding multicollinearity among predictors and its effect on ordinary least squares regression, how would the LASSO method handle multicollinearity by comparison?

It amplifies the effect of multicollinearity when compared to linear regression on the coefficient estimates.

☐ It amplifies the effect of multicollinearity when compared to linear regression on the coefficient estimates.

☐ It has no mechanism to handle multicollinearity, leading to unstable coefficients.

☒ It can handle multicollinearity better because it shrinks coefficients of correlated predictors, sometimes to zero.

☐ It ignores multicollinearity entirely and is completely unaffected when compared to linear regression.

Correct

Correct: LASSO can deal with multicollinearity by shrinking the coefficients, potentially setting some to zero, thus reducing the variance of the estimates.

10.

In LASSO regression, what is the impact of the lambda penalty as it approaches zero?

The coefficients of the predictors are increasingly penalized and shrink to zero.

☐ The coefficients of the predictors are increasingly penalized and shrink to zero.

☐ The model becomes equivalent to a Ridge Regression model.

☒ The LASSO regression model becomes similar to an ordinary least squares regression model.

☐ The number of predictors selected by the model decreases.

Correct

11.

In which method is the risk of overfitting generally considered to be lower compared to best subset selection?

Ridge regression

☐ Ridge regression

☐ Least squares method

☒ Forward stepwise selection

☐ Best subset selection

Correct

12.

In the context of subset selection, what is a potential disadvantage of using forward stepwise selection over best subset selection?

It evaluates too many models, leading to a high risk of overfitting.

☐ It evaluates too many models, leading to a high risk of overfitting.

☒ It might not include predictors that would be identified in a more exhaustive search.

☐ It requires that the number of predictors be less than the number of observations.

☐ It is more computationally intensive than best subset selection.

Correct

13.

What effect does multicollinearity have on the estimates produced by a linear regression model?

It increases the overall fit of the model as measured by R-squared.

☐ It increases the overall fit of the model as measured by R-squared.

☐ It leads to a decrease in the standard error of the regression coefficients.

☒ It can make the coefficient estimates unstable.

☐ It reduces the computational time required to fit the model.

Correct

14.

Why might a data scientist apply a square root transformation to a predictor in a linear regression model?

To reduce the effect of skewness and extreme values in the predictor.

☒ To reduce the effect of skewness and extreme values in the predictor.

☐ To ensure that all regression coefficients will be squared.

☐ To allow the regression model to fit polynomial relationships.

☐ To increase the linearity of the relationship between predictor and response.

Correct

15.

What is the result of applying a differencing transformation to a time-series data in the context of linear regression?

It helps to remove trends and seasonality from the data.

☒ It helps to remove trends and seasonality from the data.

☐ It ensures that the future values of the series can be predicted without error.

☐ It changes the time intervals of the data collection.

☐ It converts the series into a stationary series.

Correct

https://www.coursera.org/learn/linois-tech-statistical-learning/assignment-submission/lieh/module-2-summative-assessment/view-feedback

1/1